

OL7014 Integer Programming

Homework 1 — Solutions

Due Sep 15, start of class

Write down your solutions and submit them as a single PDF to loss0f0706@gmail.com.
 Problem 6 is an optional bonus (20 points).

Problem 1. Reading and writing the language

(20 points)

A transport aircraft is being loaded for a resupply flight. Three cargo types:

	water pallet	fuel drum	medical pallet
value (pts)	5	4	7
weight (t)	2	3	1
volume (m ³)	3	1	2

The aircraft carries at most 60 t and 55 m³. Orders require that *exactly* 4 medical pallets fly. For this problem cargo is divisible — fractional pallets may be manifested. The goal is to maximize total value.

- (a) **(6 pts)** Define the decision variables and write the problem as a scalar LP: objective plus constraints, one line each, with a few words saying what each constraint enforces.

Solution. Let x_1, x_2, x_3 be the numbers of water pallets, fuel drums, and medical pallets loaded (fractions allowed).

$$\begin{array}{ll}
 \max & 5x_1 + 4x_2 + 7x_3 \quad \text{total value} \\
 \text{s.t.} & 2x_1 + 3x_2 + x_3 \leq 60 \quad \text{weight limit (t)} \\
 & 3x_1 + x_2 + 2x_3 \leq 55 \quad \text{volume limit (m}^3\text{)} \\
 & x_3 = 4 \quad \text{exactly four medical pallets fly} \\
 & x_1, x_2, x_3 \geq 0 \quad \text{no negative cargo}
 \end{array}$$

- (b) **(8 pts)** Put your LP in the standard form $\max c^\top x$ s.t. $Ax \leq b$ from Week 1. Write out c , A , b with numbers in them. Remember the dictionary: an equality becomes *two* inequality rows, and nonnegativity gets its own rows. Report the dimensions of A .

Solution. With $x = (x_1, x_2, x_3)^\top$: the equality $x_3 = 4$ becomes the pair $x_3 \leq 4$, $-x_3 \leq -4$, and each sign constraint $x_j \geq 0$ becomes $-x_j \leq 0$.

$$c = \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix}, \quad A = \begin{pmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 60 \\ 55 \\ 4 \\ -4 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{array}{l} \text{weight} \\ \text{volume} \\ x_3 \leq 4 \\ x_3 \geq 4 \\ x_1 \geq 0 \\ x_2 \geq 0 \\ x_3 \geq 0 \end{array}$$

A is 7×3 : seven constraint rows, three variables.

- (c) **(3 pts)** Is the feasible set a polyhedron? Is the problem convex? One sentence each.

Solution. Yes: the feasible set is $\{x : Ax \leq b\}$, the intersection of finitely many half-spaces, which is a polyhedron by definition. Yes: a polyhedron is a convex set and the objective is linear, so the problem is convex.

- (d) **(3 pts)** Orders change: only *whole* pallets and drums may fly ($x \in \mathbb{Z}^3$). Which of your answers in (c) change, and to what?

Solution. Both flip to *no*. The feasible set is now $\{x \in \mathbb{Z}^3 : Ax \leq b\}$, a finite set of isolated lattice points: it is not convex — $(0, 0, 4)$ and $(1, 0, 4)$ are both feasible, but their midpoint $(\frac{1}{2}, 0, 4)$ is not — and since every polyhedron is convex, it is not a polyhedron either. A non-convex feasible set makes the problem non-convex.

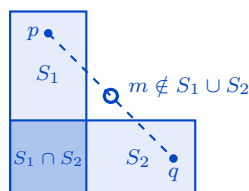
Problem 2. Convexity, by pictures

(20 points)

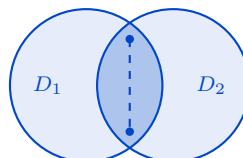
No formal proofs in this problem. All you need is the *segment test* from lecture — a set is convex when the straight segment between any two of its points stays inside — plus a pencil and a ruler. Label your drawings.

- (a) (7 pts) Draw two convex sets in $\mathbb{R}^2$ whose *union* is not convex: mark two points of the union and the point on their segment that escapes it. Then *try* to draw two convex sets whose *intersection* is not convex. After a few honest attempts, state in one sentence what you believe to be true about intersections — no proof needed, just your pictures.

Solution.



union of two rectangles: **not convex**

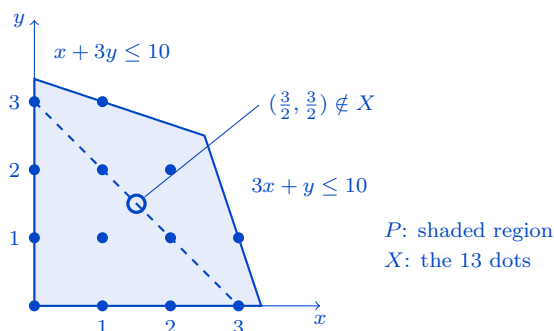


$D_1 \cap D_2$ (the lens): convex

Left: the rectangles S_1 (tall) and S_2 (wide) are convex, but their L-shaped union is not — $p \in S_1$ and $q \in S_2$, yet the midpoint m of their segment lies in neither. Every attempt at a non-convex intersection fails: the same two rectangles meet in a square, and two overlapping disks meet in a lens (right), both convex. What the pictures suggest: *the intersection of two convex sets is always convex*.

- (b) (7 pts) Draw the Week 1 toy $P = \{(x, y) \geq 0 : x + 3y \leq 10, 3x + y \leq 10\}$ and mark all the lattice points $X = P \cap \mathbb{Z}^2$. Now run the segment test on X : pick two points of X whose segment passes through a point that is not in X , and label that escaping point with its coordinates. What did the single line “ $\in \mathbb{Z}$ ” do to convexity?

Solution.



The points $(0, 3)$ and $(3, 0)$ are both in X , but the midpoint of their segment, $(\frac{3}{2}, \frac{3}{2})$, is not an integer point, so it is not in X . The line “ $\in \mathbb{Z}$ ” destroyed convexity: P is convex, but $X = P \cap \mathbb{Z}^2$ is not.

- (c) (6 pts) Same picture, one question: for a max problem, why can the LP relaxation’s optimal value z_{LP} never be *smaller* than the IP’s z_{IP} ? Answer in one or two sentences, no notation needed. (Lecture’s hint: deleting the “ $\in \mathbb{Z}$ ” line only *enlarges* the set of allowed plans.)

Solution. Every plan the IP allows (each dot) is also allowed by the LP (it lies in the shaded region), so the LP picks its best plan from a larger menu that contains the IP’s best plan — it can match that plan, so it can never do worse.

Problem 3. Relax, round, regret

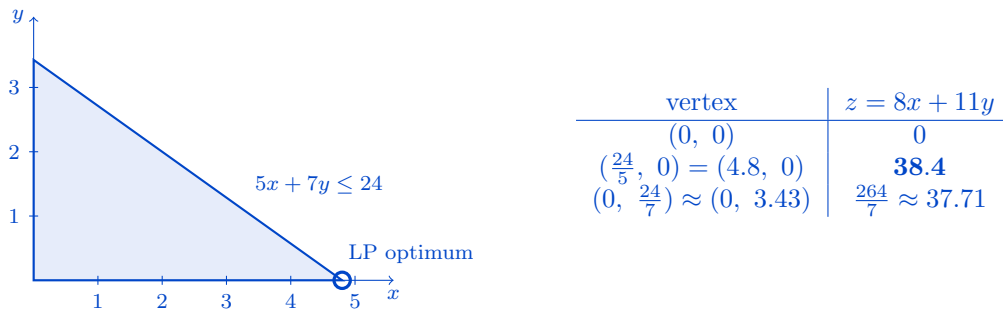
(20 points)

A patrol packs a rucksack that carries $W = 24$ kg. Ammo boxes weigh 5 kg and are worth 8 points each; med kits weigh 7 kg and are worth 11 points each. Any nonnegative integer count of each may be packed:

$$\begin{aligned} \max \quad & 8x + 11y \\ \text{s.t.} \quad & 5x + 7y \leq 24 \\ & x, y \in \mathbb{Z}_+ \end{aligned}$$

- (a) (5 pts) Solve the LP relaxation graphically: draw the feasible region, list every vertex with its objective value, and report the LP optimum (x^*, y^*) and z_{LP} .

Solution. The relaxation keeps $5x + 7y \leq 24$, $x, y \geq 0$: a triangle.



LP optimum: $(x^*, y^*) = (4.8, 0)$, $z_{\text{LP}} = 38.4$.

- (b) (5 pts) Round the LP optimum: list *every* candidate obtained by taking $\lfloor \cdot \rfloor$ or $\lceil \cdot \rceil$ in each coordinate, check each for feasibility in a table (as in lecture), and report the best *feasible* rounded value.

Solution. In x : $\lfloor 4.8 \rfloor = 4$ or $\lceil 4.8 \rceil = 5$. In y : $y^* = 0$ is already an integer, so $\lfloor 0 \rfloor = \lceil 0 \rceil = 0$. The $2^2 = 4$ combinations collapse to two distinct candidates:

candidate	$5x + 7y \leq 24$	$z = 8x + 11y$
$(4, 0)$	20 ✓	32
$(5, 0)$	25 ×	infeasible

Best feasible rounding: $(4, 0)$ with value 32.

- (c) (5 pts) Find the true IP optimum by enumeration. First justify why $x \leq 4$ and $y \leq 3$ bound the search, then report the optimum and its value.

Solution. Since $y \geq 0$, we have $5x \leq 5x + 7y \leq 24$, so $x \leq 4.8$, and x is an integer, so $x \leq 4$. Likewise $7y \leq 24$ gives $y \leq 3.43$, so $y \leq 3$. Every feasible plan therefore lies in the 5×4 grid $\{0, \dots, 4\} \times \{0, \dots, 3\}$. Its objective values ($\times$ = overweight):

	$x = 0$	$x = 1$	$x = 2$	$x = 3$	$x = 4$
$y = 0$	0	8	16	24	32
$y = 1$	11	19	27	35	×
$y = 2$	22	30	38	×	×
$y = 3$	33	×	×	×	×

IP optimum: $(x, y) = (2, 2)$, $z_{\text{IP}} = 38$.

- (d) (2 pts) Verify the chain $z_{\text{LP}} \geq z_{\text{IP}} >$ best rounding with your numbers, and say in one sentence what rounding lost.

Solution. $z_{\text{LP}} = 38.4 \geq z_{\text{IP}} = 38 > 32$ ✓. Rounding can only move to grid points next to $(4.8, 0)$, so it never considers a single med kit and leaves 4 kg unused, while the optimum trades two ammo boxes for two med kits, fills the rucksack exactly, and scores 6 points more.

- (e) (3 pts) Implement the IP with `gurobipy` and solve it. Include a screenshot of your code and its printed output, using exactly this form:

```
print(f"optimal solution x is {x.X}")
print(f"optimal solution y is {y.X}")
print(f"optimal objective value is {model.ObjVal}")
```

Check that Gurobi agrees with your answer in (c).

Solution. Code:

```
import gurobipy as gp
from gurobipy import GRB

model = gp.Model("rucksack")
x = model.addVar(vtype=GRB.INTEGER, lb=0, name="x")    # ammo boxes
y = model.addVar(vtype=GRB.INTEGER, lb=0, name="y")    # med kits
model.setObjective(8*x + 11*y, GRB.MAXIMIZE)
model.addConstr(5*x + 7*y <= 24, name="weight")
model.optimize()

print(f"optimal solution x is {x.X}")
print(f"optimal solution y is {y.X}")
print(f"optimal objective value is {model.ObjVal}")
```

Printed output:

```
optimal solution x is 2.0
optimal solution y is 2.0
optimal objective value is 38.0
```

Gurobi returns (2, 2) with value 38, in agreement with (c).

Problem 4. The rewriting dictionary

(20 points)

- (a) (10 pts) Convert the following problem to the standard form $\max \tilde{c}^\top \tilde{x}$ s.t. $\tilde{A}\tilde{x} \leq \tilde{b}$:

$$\begin{aligned} \min \quad & 2x_1 - x_2 + 3x_3 \\ \text{s.t.} \quad & x_1 + x_2 \geq 4 \\ & 2x_1 - x_3 = 5 \\ & x_2 + x_3 \leq 6 \\ & x_1 \geq 0, \quad x_3 \geq 0, \quad x_2 \text{ free} \end{aligned}$$

List the new variable vector $\tilde{x}$, write out $\tilde{c}$, $\tilde{A}$, $\tilde{b}$ with numbers, report the dimensions of $\tilde{A}$, and state how the optimal value of the rewritten problem relates to the original one. *Follow-up:* if x_2 were declared integer, which of your new variables must be declared integer — and show by example why declaring only one of them integer is wrong.

Solution. Four dictionary rules: split the free variable, $x_2 = x_2^+ - x_2^-$ with $x_2^+, x_2^- \geq 0$; turn min into max of the negative; flip the $\geq$ row; replace the $=$ row by two rows. The new variable vector is $\tilde{x} = (x_1, x_2^+, x_2^-, x_3)^\top \in \mathbb{R}^4$, and the objective becomes $\min(2x_1 - x_2^+ + x_2^- + 3x_3) = -\max(-2x_1 + x_2^+ - x_2^- - 3x_3)$:

$$\tilde{c} = \begin{pmatrix} -2 \\ 1 \\ -1 \\ -3 \end{pmatrix}, \quad \tilde{A} = \begin{pmatrix} -1 & -1 & 1 & 0 \\ 2 & 0 & 0 & -1 \\ -2 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} -4 \\ 5 \\ -5 \\ 6 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{matrix} x_1 + x_2 \geq 4 \\ 2x_1 - x_3 \leq 5 \\ 2x_1 - x_3 \geq 5 \\ x_2 + x_3 \leq 6 \\ x_1 \geq 0 \\ x_2^+ \geq 0 \\ x_2^- \geq 0 \\ x_3 \geq 0 \end{matrix}$$

$\tilde{A}$ is 8×4 .

Optimal values: the two problems have the same optimal plans, with $x_2 = x_2^+ - x_2^-$, and opposite optimal values: original optimum = $-(\text{rewritten optimum})$.

Follow-up: both x_2^+ and x_2^- must be declared integer (x_1 and x_3 stay continuous). Suppose only x_2^+ is declared integer. Then $\tilde{x} = (3, 5, 0.5, 1)$ satisfies all eight rows ($x_1 + x_2 = 7.5 \geq 4$, $2 \cdot 3 - 1 = 5$, $x_2 + x_3 = 5.5 \leq 6$) and has $x_2^+ = 5$ integer, yet it encodes $x_2 = 5 - 0.5 = 4.5$, a plan the integer model forbids. Symmetrically, if only x_2^- is integer, $(x_2^+, x_2^-) = (4.5, 0)$ gets in.

- (b) **(10 pts)** Ten escort vehicles must be split across two convoy routes: $x_1 + x_2 = 10$, $x \geq 0$. The mission's exposure is the *worst* of three segment scores, each decreasing in the escorts assigned:

$$s_1 = 40 - 3x_1 - x_2, \quad s_2 = 38 - x_1 - 4x_2, \quad s_3 = 36 - 2x_1 - 2x_2.$$

Command wants to minimize the worst exposure: $\min_x \max\{s_1, s_2, s_3\}$.

- (i) **(5 pts)** Reformulate as an LP using the epigraph trick. Write out the full LP.

Solution. Add one free variable t that must sit on or above every score:

$$\begin{aligned} \min_{x_1, x_2, t} \quad & t \\ \text{s.t.} \quad & 40 - 3x_1 - x_2 \leq t \\ & 38 - x_1 - 4x_2 \leq t \\ & 36 - 2x_1 - 2x_2 \leq t \\ & x_1 + x_2 = 10 \\ & x_1, x_2 \geq 0, \quad t \text{ free} \end{aligned}$$

- (ii) **(2 pts)** In one sentence: why does the optimal t equal the largest s_i at the optimum, rather than sitting strictly above it?

Solution. The rows $s_i \leq t$ only push t up, while the objective pulls t down, so if t sat strictly above every s_i we could lower it (keeping x), stay feasible, and improve the objective — hence at the optimum t rests on the largest s_i .

- (iii) **(3 pts)** A teammate tries the same trick on $\min_x \min_i \{s_i(x)\}$ by writing $s_i(x) \geq t \forall i$ and maximizing t . Explain why this does not model the problem, and name the Week 2–3 tool that models “the best of several cases” correctly.

Solution. The rows $s_i(x) \geq t$ make t a lower bound on all three scores, and maximizing pushes t up onto $\min_i s_i(x)$ — but maximizing also chooses x to make that minimum as *large* as possible, so the LP solves $\max_x \min_i s_i(x)$, a different problem. Switching to *minimizing* t does not rescue it: nothing then holds t up, and the LP is unbounded. A min of a min needs $t \geq \min_i s_i(x)$, which says “ $t \geq s_1$ or $t \geq s_2$ or $t \geq s_3$ ”: a union of half-spaces, which is not convex, so no LP can describe it.

The right tool is the **either–or** construction (“at least k of m constraints” with $k = 1$): one binary switch z_i per case,

$$\begin{aligned} \min_{x, t, z} \quad & t \\ \text{s.t.} \quad & s_i(x) \leq t + M(1 - z_i) \quad (i = 1, 2, 3), \quad z_1 + z_2 + z_3 \geq 1, \\ & x_1 + x_2 = 10, \quad x \geq 0, \quad t \text{ free}, \quad z \in \{0, 1\}^3. \end{aligned}$$

Here $M = 32$ is valid: it is the largest gap between two scores on the domain.

Problem 5. Switches

(20 points)

Four candidate sites for forward supply depots must together ship 100 t to the front. Opening site i costs c_i ; an open site ships $y_i \geq 0$ tons at f_i per ton, up to its capacity u_i ; a closed site ships nothing.

site i	1 (island)	2 (port)	3 (north road)	4 (south road)
fixed cost c_i (\$k)	18	12	16	14
capacity u_i (t)	80	50	70	40
per-ton cost f_i (\$k/t)	0.2	0.5	0.3	0.25

- (a) (6 pts) Write the base MILP: binaries x_i for opening, continuous y_i for shipping, objective, demand constraint, and one *linking constraint* per site. Why is u_i the smallest certainly-valid big- M in site i 's linking row?

Solution. $x_i = 1$ if site i opens; $y_i =$ tons shipped from site i .

$$\begin{array}{ll} \min & 18x_1 + 12x_2 + 16x_3 + 14x_4 + 0.2y_1 + 0.5y_2 + 0.3y_3 + 0.25y_4 \quad \text{fixed + shipping cost (\$k)} \\ \text{s.t.} & y_1 + y_2 + y_3 + y_4 = 100 \quad \text{demand} \\ & y_1 \leq 80x_1, \quad y_2 \leq 50x_2, \quad y_3 \leq 70x_3, \quad y_4 \leq 40x_4 \quad \text{linking, } y_i \leq u_i x_i \\ & x \in \{0, 1\}^4, \quad y \geq 0 \end{array}$$

Why $M = u_i$: when $x_i = 1$ the row must not cut off any shipment an open site can really make, so M must be at least the largest possible y_i , which is the capacity u_i (the demand of 100 t exceeds every capacity, so it gives no tighter cap). Any $M < u_i$ would forbid legal shipments between M and u_i .

- (b) (6 pts) Add each field rule as one or two linear constraints, and check each by trying the 0/1 cases:
- (i) site 1 is on an island supplied through the port — if site 1 opens, site 2 must open;
 - (ii) sites 3 and 4 share one mountain road — at most one of them may open;
 - (iii) command requires at least two open sites.

Solution. One constraint each:

rule	constraint	0/1 cases kept	0/1 cases cut
(i)	$x_1 \leq x_2$	$(x_1, x_2) = (0, 0), (0, 1), (1, 1)$	$(1, 0)$: $1 \leq 0$ fails
(ii)	$x_3 + x_4 \leq 1$	$(x_3, x_4) = (0, 0), (1, 0), (0, 1)$	$(1, 1)$: $2 \leq 1$ fails
(iii)	$x_1 + x_2 + x_3 + x_4 \geq 2$	≥ 2 open sites (11 plans)	0 or 1 open site (5 plans)

In each case the constraint cuts exactly the forbidden cases and nothing else.

- (c) (3 pts) A new rule arrives: if *any single site* ships more than 60 t, an inspector must be hired for a flat \$5k. Model this with one new binary z and one constraint per site. Give the smallest certainly-valid M for each of the four rows — and point out which rows turn out to be vacuous, and why.

Solution. Let $z = 1$ if the inspector is hired. Add $5z$ to the objective and the rows

$$y_i \leq 60 + M_i z \quad (i = 1, 2, 3, 4).$$

If $z = 0$, every site ships at most 60 t; if $z = 1$, the rows switch off. The smallest valid M_i is the largest possible y_i minus 60, and the linking row caps y_i at u_i :

site i	1	2	3	4
$u_i - 60$	20	-10	10	-20
smallest valid M_i	20	0 (vacuous)	10	0 (vacuous)

Rows 2 and 4 are vacuous: those sites' capacities (50 t and 40 t) are below 60 t, so $y_2 \leq 60$ and $y_4 \leq 60$ already follow from the linking rows; $M = 0$ suffices, and the two rows can be deleted.

- (d) (2 pts) Two sentences on big- M in this model: what silently goes wrong if site 1's linking M is 50, and what goes wrong (differently) if it is 10^6 ?

Solution. With $M = 50$ the model is *wrong*: $y_1 \leq 50x_1$ silently cuts the island's capacity from 80 t to 50 t, and the solver confidently returns the best plan of the wrong problem, with no warning. With $M = 10^6$ the model stays valid (once $y_1 \leq 80$ is written as its own row, since the linking row was doing the capacity's job), but it is *weak*: a tiny fractional x_1 already unlocks a large shipment while paying almost none of the fixed cost, so the LP relaxation bound is poor and the solver must branch much more to prove optimality.

- (e) (3 pts) Solve the full model (a)–(c) with **gurobipy** in Colab. Report which sites open, the shipments y , whether the inspector is hired, and the total cost. Then delete the inspector rule, re-solve, and report what changes.

Solution.

```

import gurobipy as gp
from gurobipy import GRB

c = [18, 12, 16, 14]; u = [80, 50, 70, 40]; f = [0.2, 0.5, 0.3, 0.25]
S = range(4)

m = gp.Model("depots")
x = m.addVars(S, vtype=GRB.BINARY, name="x")    # open site i
y = m.addVars(S, lb=0, name="y")                # tons shipped from site i
z = m.addVar(vtype=GRB.BINARY, name="z")        # inspector hired
m.setObjective(gp.quicksum(c[i]*x[i] + f[i]*y[i] for i in S) + 5*z, GRB.MINIMIZE)
m.addConstr(y.sum() == 100, name="demand")
m.addConstrs((y[i] <= u[i]*x[i] for i in S), name="link")
m.addConstr(x[0] <= x[1], name="island_needs_port")
m.addConstr(x[2] + x[3] <= 1, name="mountain_road")
m.addConstr(x.sum() >= 2, name="two_sites")
m.addConstr(y[0] <= 60 + 20*z, name="inspect_1")
m.addConstr(y[2] <= 60 + 10*z, name="inspect_3")
m.optimize()
print("open:", [i+1 for i in S if x[i].X > 0.5])
print("y:", [y[i].X for i in S], " inspector:", round(z.X), " cost:", m.ObjVal)

# delete the inspector rule and re-solve
m.remove([m.getConstrByName("inspect_1"), m.getConstrByName("inspect_3"), z])
m.optimize()
print("open:", [i+1 for i in S if x[i].X > 0.5])
print("y:", [y[i].X for i in S], " cost:", m.ObjVal)

```

Printed output:

```

open: [1, 2]
y: [80.0, 20.0, 0.0, 0.0]  inspector: 1  cost: 61.0
open: [1, 2]
y: [80.0, 20.0, 0.0, 0.0]  cost: 56.0

```

Full model: sites 1 (island) and 2 (port) open; $y = (80, 20, 0, 0)$ t; the inspector *is* hired; total cost \$61k. **Without the inspector rule:** the same sites and the same shipments at \$56k — the only change is the \$5k fee.

Problem 6. Standard form, without understanding (+20 bonus points)

The model below is the *deterministic location-allocation problem* (LAP); it returns in Weeks 5 and 9–10. You are **not** expected to understand the problem — just apply the Week 2 dictionary.

Sites $i \in \{1, 2, 3\}$; directed arcs $a_1 = (1 \rightarrow 2)$, $a_2 = (2 \rightarrow 1)$, $a_3 = (2 \rightarrow 3)$, $a_4 = (3 \rightarrow 2)$. Decisions: open $x_i \in \{0, 1\}$, reserve $r_i \geq 0$, flow $w_a \geq 0$, shortage $q_i \geq 0$, surplus $s_i \geq 0$.

$$\begin{aligned}
 \min_{x, r, w, q, s} \quad & \sum_i c_i x_i + \sum_i f_i r_i + \sum_a t_a w_a + \pi \sum_i q_i + \sigma \sum_i s_i \\
 \text{s.t.} \quad & \sum_i r_i \leq B && \text{(budget)} \\
 & r_i \leq B x_i \quad \forall i && \text{(linking)} \\
 & \sum_i x_i \leq p && \text{(cardinality)} \\
 & \sum_{a \text{ into } i} w_a - \sum_{a \text{ out of } i} w_a + q_i - s_i + r_i = \bar{\zeta}_i \quad \forall i && \text{(balance)} \\
 & 0 \leq w_a \leq w^{\max} \quad \forall a, \quad 0 \leq q_i \leq \bar{\zeta}_i \quad \forall i, \quad r, s \geq 0, \quad x_i \in \{0, 1\} \quad \forall i
 \end{aligned}$$

site i	1	2	3	arc a	a_1	a_2	a_3	a_4
open cost c_i	100	80	120	transport t_a	4	4	6	6
reserve cost f_i	2	1	3					
mean demand $\bar{\zeta}_i$	12	8	15					

$\pi = 60$ (shortage), $\sigma = 1$ (surplus), $B = 40$, $p = 2$, $w^{\max} = 90$

Reduce everything to the course's canonical MILP form (Week 2, Section 1):

$$\max c^\top x + h^\top y \quad \text{s.t.} \quad Ax + Gy \leq b, \quad x \in \mathbb{Z}_+^3, \quad y \geq 0$$

— remember the convention: x = integer decisions, y = continuous.

- (a) **(4 pts)** Which decisions form the integer block x , and which the continuous block y ? Write y as a stacked vector in the order (r, w, q, s) , report the dimensions of x and y , and give the dictionary line that turns each $x_i \in \{0, 1\}$ into the form above.

Solution. Integer block: the opening decisions, $x = (x_1, x_2, x_3)^\top \in \mathbb{Z}_+^3$, dimension 3. Continuous block: everything else, stacked as

$$y = (r_1, r_2, r_3, w_{a_1}, w_{a_2}, w_{a_3}, w_{a_4}, q_1, q_2, q_3, s_1, s_2, s_3)^\top \in \mathbb{R}^{13}, \quad 3 + 4 + 3 + 3 = 13.$$

Dictionary line: $x_i \in \{0, 1\}$ becomes $x_i \in \mathbb{Z}$, $0 \leq x_i \leq 1$. Integrality and $x_i \geq 0$ are already part of $x \in \mathbb{Z}_+^3$; the upper bound becomes one ordinary row $x_i \leq 1$ per site, three rows in all.

- (b) **(5 pts)** Write c and h in full, with numbers. (Careful: the model is a min.)

Solution. By the first dictionary line, $\min(\text{cost}) = -\max(-\text{cost})$, so every cost coefficient changes sign:

$$c = (-100, -80, -120)^\top, \quad h = (\underbrace{-2, -1, -3}_r, \underbrace{-4, -4, -6, -6}_w, \underbrace{-60, -60, -60}_q, \underbrace{-1, -1, -1}_s)^\top.$$

- (c) **(9 pts)** Describe A , G , b as labeled blocks with their dimensions (budget, linking, cardinality, balance, bounds). Then write out, in full numbers — each row of A together with its row of G — these rows only: the budget row, the three linking rows, the cardinality row, and the *two* rows that encode site 2's balance equality.

Solution. Split the columns of G as $[r \mid w \mid q \mid s]$, of widths $3 \mid 4 \mid 3 \mid 3$. Write I_k for the identity, $\mathbf{1}$ for an all-ones vector, and 0 for a zero block of the right size. The flows enter the balance rows through the 3×4 *node-arc matrix* N (+1 if arc a enters site i , -1 if it leaves), and the linking rows are moved into the form $r_i - 40x_i \leq 0$:

$$N = \begin{pmatrix} -1 & 1 & 0 & 0 \\ 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{pmatrix} \quad \begin{matrix} \text{site 1} \\ \text{site 2} \\ \text{site 3} \end{matrix} \quad (\text{columns } a_1, a_2, a_3, a_4)$$

block	rows	A ($\cdot \times 3$)	G ($\cdot \times 13$)	b
budget	1	0	$[\mathbf{1}^\top \mid 0 \mid 0 \mid 0]$	40
linking	3	$-40 I_3$	$[I_3 \mid 0 \mid 0 \mid 0]$	0
cardinality	1	$\mathbf{1}^\top$	0	2
balance, “ $\leq$ ” half	3	0	$[I_3 \mid N \mid I_3 \mid -I_3]$	$\bar{\zeta} = (12, 8, 15)^\top$
balance, “ $\geq$ ” half, flipped	3	0	$[-I_3 \mid -N \mid -I_3 \mid I_3]$	$-\bar{\zeta}$
bounds $x_i \leq 1$	3	I_3	0	$\mathbf{1}$
bounds $w_a \leq w^{\max}$	4	0	$[0 \mid I_4 \mid 0 \mid 0]$	$90 \cdot \mathbf{1}$
bounds $q_i \leq \bar{\zeta}_i$	3	0	$[0 \mid 0 \mid I_3 \mid 0]$	$\bar{\zeta}$
total	21	21×3	21×13	21 entries

The requested rows in full, with each row of A beside its row of G (every row reads “ $\leq b$ ”):

	A			G													b
	x_1	x_2	x_3	r_1	r_2	r_3	w_{a_1}	w_{a_2}	w_{a_3}	w_{a_4}	q_1	q_2	q_3	s_1	s_2	s_3	
budget	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	40
linking 1	-40	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
linking 2	0	-40	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
linking 3	0	0	-40	0	0	1	0	0	0	0	0	0	0	0	0	0	0
cardinality	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	2
site 2, $\leq$	0	0	0	0	1	0	1	-1	-1	1	0	1	0	0	-1	0	8
site 2, $\geq$	0	0	0	0	-1	0	-1	1	1	-1	0	-1	0	0	1	0	-8

Site 2's balance reads $w_{a_1} + w_{a_4} - w_{a_2} - w_{a_3} + q_2 - s_2 + r_2 = 8$: arcs a_1 and a_4 come in, a_2 and a_3 go out. The first row is its “ $\leq$ ” half, and the second is the “ $\geq$ ” half multiplied by -1.

- (d) **(2 pts)** Report the final dimensions of A and G , counting every row your conversion produced. (Non-negativity rows are free here — $x \in \mathbb{Z}_+^3$ and $y \geq 0$ are part of the form.)

Solution. $1 + 3 + 1 + 6 + 3 + 4 + 3 = 21$ rows (budget, linking, cardinality, balance, $x_i \leq 1$, flow bounds, shortage bounds), so A is 21×3 , G is 21×13 , and $b \in \mathbb{R}^{21}$.